

Tien Pham

(Pham Canh An Tien)

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Web: Portfolio

EDUCATION

- University of Manchester** **Manchester, UK.**
PhD student in Computer Science October 2023 - September 2026
Interactive Language Explanation and Visualization for Explainable Robotics
Advisor: Angelo Cangelosi, Ph.D.
Research Interest: Explainable Reinforcement Learning, VLM, Robotics
- University of Texas at Arlington** **Arlington, TX, U.S.**
Master of Science in Computer Science January 2021 - June 2023
GPA: 4.00/4.00
Thesis: Context-Aware Gaze-Based Interface and Smart Wheelchair
Advisor: Manfred Huber, Ph.D.
- University of Technology, Vietnam National University** **Ho Chi Minh City, Vietnam**
Bachelor of Engineering in Mechatronics (PFIEV Program) July 2014 - July 2019
Addendum to the Engineering Diploma of The Program of Training Engineers of Excellence in Vietnam
GPA: 8.21/10.00
Thesis: Design of Undulating Fin Applying for Autonomous Underwater Vehicle (*Score:* 9.54/10.00)
Advisor: Tan-Tien Nguyen, Ph.D.

TEACHING EXPERIENCE

- Graduate Teaching Assistant, COMP34212: *Cognitive Robotics*, 2024, 2025, 2026, UoM.
- Graduate Teaching Assistant, COMP23311: *Software Engineering*, Fall 2023, UoM.
- Graduate Teaching Assistant, CSE 5301: *Data Analysis and Modeling Techniques*, Fall 2022, Spring 2023, UTA.
- Graduate Teaching Assistant, CSE 6331-004: *Advanced Topics in Database Systems*, Summer 2022, UTA.
- Graduate Teaching Assistant, CSE 3320-001: *Operating Systems*, Summer 2021, UTA.

RESEARCH EXPERIENCE

- Cognitive Robotics Lab** **Manchester, UK**
Marie Curie Early Career Researcher October 2023 - September 2026
 - Study on Explainable Deep Reinforcement Learning, Vision Language Models (VLMs) for Generalizable, Explainable, Trustworthy, Transparent and Interpretable Robot.
 - Skills:* Computer Vision, Vision Language Models (VLMs), Task And Motion Planning (TAMP), Explainable Reinforcement Learning, Attention Mechanism, Human Robot Interaction.
- Learning and Adaptive Robotics (LEARN) Lab** **Arlington, TX, U.S.**
Graduate Research Assistant September 2022 - June 2023
 - Developing the Context-Aware Interface for disabled to interact with the wheelchair hand-freely while implement autonomous control for safety assurance.
 - Skills:* Computer Vision, ROS, C++, Head pose Estimation, Eye gazing estimation, Object Detection, Newton Gaussian Method, MATLAB
- Wireless and Sensor Systems Lab** **Arlington, TX, U.S.**
Graduate Research Assistant January 2021 - May 2022
 - Worked on wearable devices for Medical Application such as FaceSense, BioFace, Seizure Detection on Wearable Device
 - Skills:* Signal Processing, Machine Learning, Python, Data Analysis, Convolutional Neural Network
- Robert BOSCH Engineering & Business Solutions Vietnam** **Ho Chi Minh City, Vietnam**
Embedded Software Developer September 2019 - December 2020
 - Designed and executed platform of AUTOSAR Architecture for ECUs in automotive domain.
 - Developed OSEK Network Management for integrating to AUTOSAR Architecture.
- Vietnam National DCSELab** **Ho Chi Minh City, Vietnam**
Undergraduate Research Assistant July 2016 - July 2019
 - Designed and developed the gymnotiform undulating fin module.
 - Developed the computational model simulating the process of undulating fin in CFD.
 - Modeled the forces, moment, and other external forces affecting to the swimming process of the fin.

SELECTED PUBLICATIONS

- Khang Nguyen, An T Le, **Tien Pham**, Manfred Huber, Jan Peters, Minh Nhat Vu (2025). “*FlowMP: Learning Motion Fields for Robot Planning with Conditional Flow Matching*” in Proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025).
- **Tien Pham**, Angelo Cangelosi (2025). “*Pay Attention to What and Where? Interpretable Feature Extractor in Vision-based Deep Reinforcement Learning*” in Proceedings of The 38th International Joint Conference on Neural Networks (IJCNN 2025).
- Tuan Dang, Trung Tran, Khang Nguyen, **Tien Pham**, Nhat Pham, Tam Vu, and Phuc Nguyen (2022). “*IoTree: A Battery-free Wearable System with Biocompatible Sensors for Continuous Tree Health Monitoring*” in Proceedings of the 28th Annual International Conference on Mobile Computing and Networking (MobiCom 2022).
- Vimal Kakaraparthi, Qijia Shao, Charles Carver, **Tien Pham**, Nam Bui, Phuc Nguyen, Xia Zhou, and Tam Vu (2021). “*FaceSense: Sensing Face Touch with An Ear-worn System.*” in Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (pp. 1-27).
- Yi Wu, Vimal Kakaraparthi, Zhang Li, **Tien Pham**, Jian Liu, Phuc Nguyen (2021). “*BioFace-3D: Continuous 3D Facial Reconstruction through Lightweight Single-Ear Biosensors*” in Proceedings of the 27th Annual International Conference on Mobile Computing and Networking (pp. 350-363).

HONORS AND AWARDS

- Travel grant for IJCNN 2025 conference.
- Marie Skłodowska-Curie Research Fellowship
- ACM SIGMOBILE Research Highlights 2022
- Scholarships in 10 consecutive semesters in the PFIEV program, from 2014 to 2019
- Excellent Student Award, 2018 & 2019

RESEARCH VISIT

- **CNRS-LAAS Toulouse, France.** October, 2025
- **PAL Robotics, Spain.** January, 2025

PATENTS

- **Method and Apparatus for Continuous Plant Health Monitoring Using a Battery-free System with Biocompatible Implanted Sensors. (2022)**
- **Method and Apparatus for Continuous 3D Facial Reconstruction Through Lightweight Single-ear(2021).**

SKILLS & COMPETENCIES

- **Coding:** MATLAB, Python, C, C++, Javascript and SQL.
- **Competences:** Researcher, Undergrad and Graduate Teacher, Workshop Organizer, Conference Reviewer.
- **Skills:** Machine Learning, Robotics, Computer Vision, Signal Processing, Human Robot Interaction.
- **Frameworks:** Pytorch, Mujoco, Ultralytics, Sklearn, Tensorflow, Docker, ROS noetic, ROS humble.
- **Languages:** Vietnamese (Native), English (IELTS 7.0), and French (DELFI B1).

REFERENCES

Angelo Cangelosi, PhD

Professor of Machine Learning and Robotics

Department of Computer Science

The University of Manchester, Manchester, UK

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Federico Tavella, PhD

Research Associate in Machine Learning and Robotics

Centre for Robotic Autonomy in Demanding and Long Lasting Environments (CRADLE)

The University of Manchester, Manchester, UK

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